Task 2 Report: Deep Learning Image Classification

Objective

The objective of this task was to build and evaluate a small deep learning model for image classification using TensorFlow in Google Colab. The project focused on classifying cat and dog images using a Convolutional Neural Network (CNN), while also applying image preprocessing, data augmentation, model evaluation, and model saving for inference.

Dataset

The dataset used in this project was the CIFAR-10 dataset available directly in TensorFlow through `tf.keras.datasets.cifar10.load_data()`. CIFAR-10 is a standard image classification dataset that contains 10 object categories, and for this project only the **cat** and **dog** classes were selected to create a binary image classification problem.

The dataset images are small RGB images of size 32 x 32 pixels. After filtering, the dataset was divided into training, validation, and testing sets. This setup made the project simpler to train in Google Colab while still meeting the deep learning task requirements.

Preprocessing

Several preprocessing steps were applied before training the model:

- Loaded the CIFAR-10 dataset from TensorFlow.
- Filtered only the cat and dog image classes to create a binary classifier.
- Converted class labels so that cat = 0 and dog = 1.
- Normalized image pixel values by dividing by 255.0 so that all values were in the range 0 to 1.
- Split part of the training data into a validation set for monitoring model performance during training.
- Applied data augmentation using random horizontal flip, random rotation, and random zoom to improve generalization.

These preprocessing steps helped prepare the dataset properly for deep learning training and reduced the risk of overfitting.

Model Architecture

A simple Convolutional Neural Network (CNN) was used for the classification task. The model architecture included:

- An input layer with image shape 32 x 32 x 3.
- A data augmentation layer for transforming training images.
- Three convolutional layers with ReLU activation for feature extraction.
- Two max-pooling layers for reducing spatial dimensions.
- A flatten layer to convert extracted features into a one-dimensional vector.
- A dense hidden layer with 128 neurons and ReLU activation.
- A dropout layer with 0.3 dropout rate to reduce overfitting.
- A final output layer with sigmoid activation for binary classification.

The model was compiled using the Adam optimizer, binary crossentropy loss function, and accuracy as the main evaluation metric.

Results

The model was trained for 5 epochs with a batch size of 32. During training, both training accuracy/loss and validation accuracy/loss were plotted to observe learning progress across epochs.

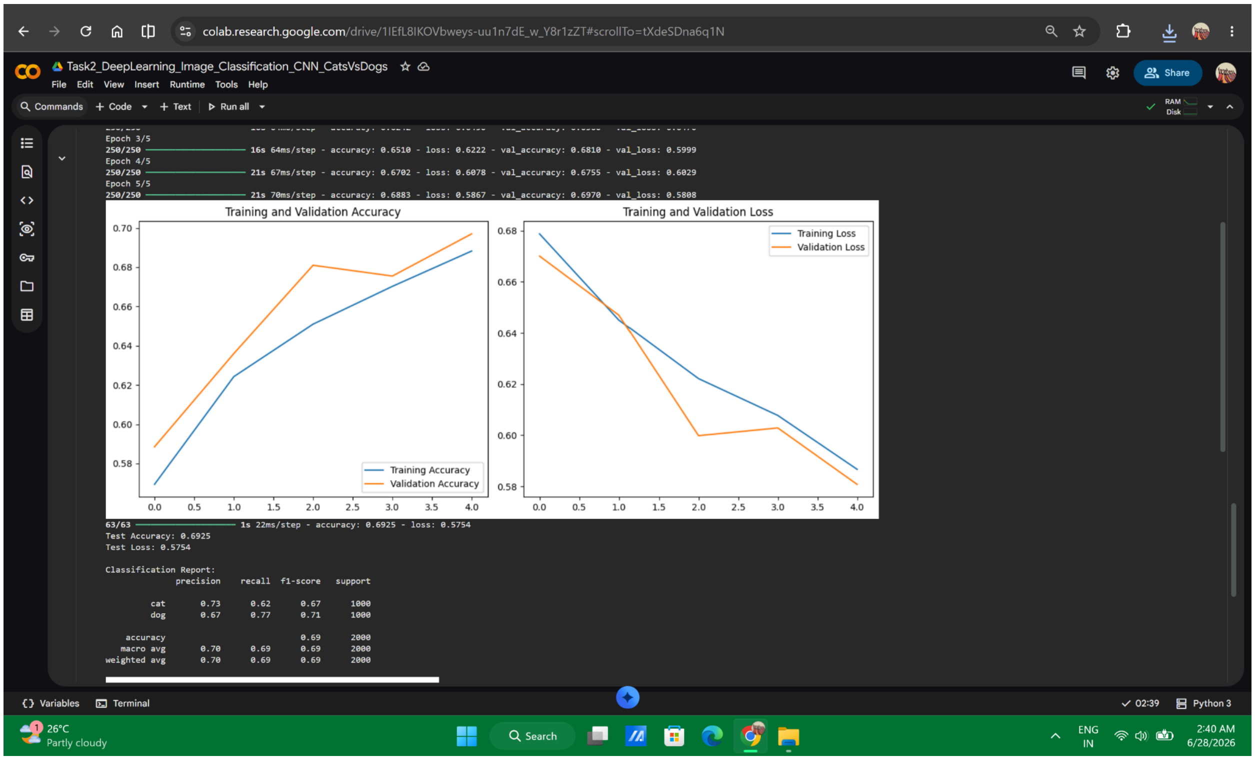
After training, the model was evaluated on the test dataset and produced test accuracy and test loss values. Additional evaluation was performed using a classification report and confusion matrix, which provided a clearer understanding of the model's performance on cat and dog image classification.

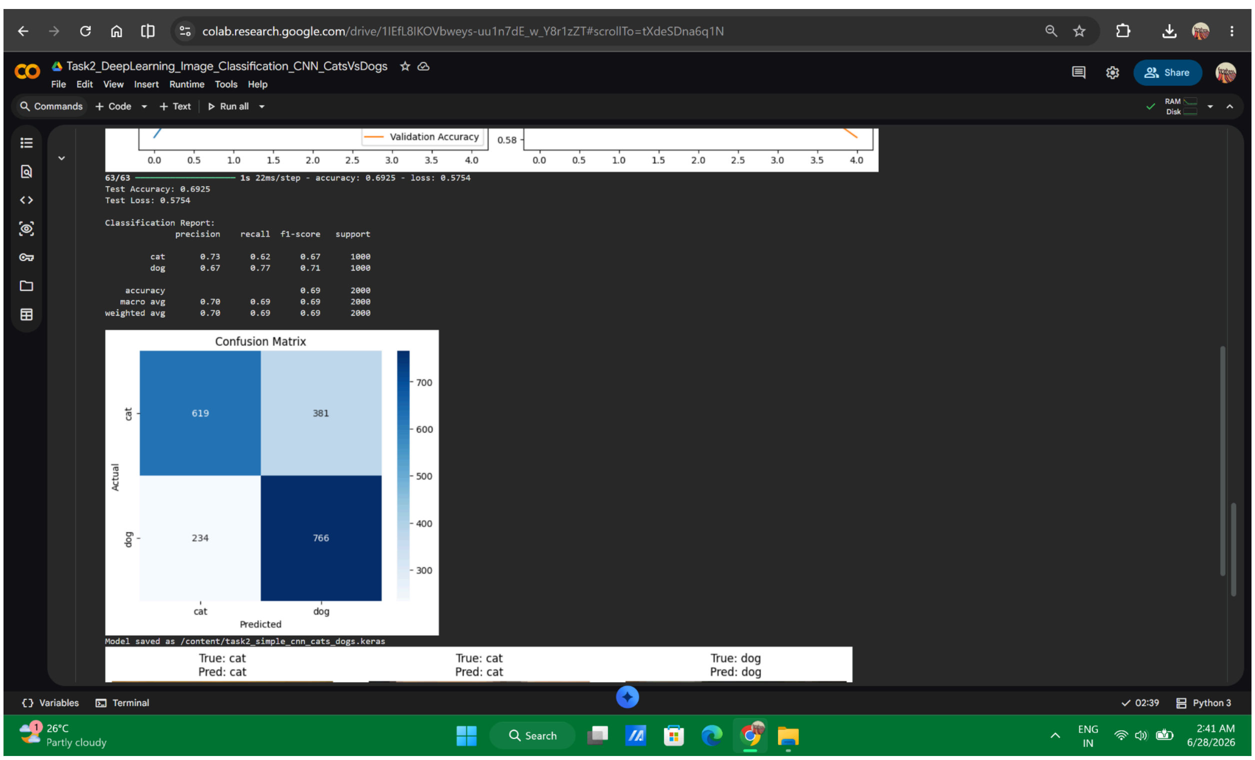
The notebook also displayed sample predictions by comparing true labels and predicted labels on test images, which made the output easier to interpret visually. Finally, the trained model was saved as a `.keras` file for later reuse and inference.

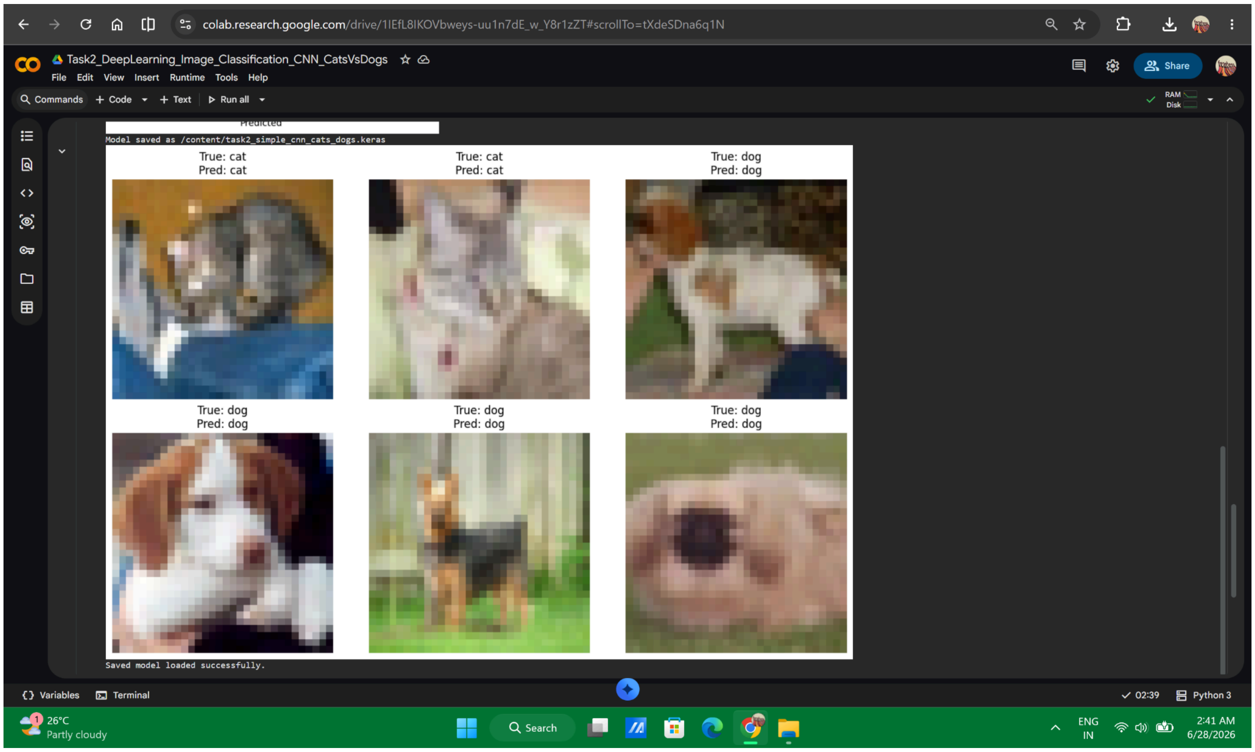
Conclusion

This project successfully demonstrated the implementation of a small deep learning image classification model in Google Colab using TensorFlow. The task provided practical experience in loading image datasets, preprocessing images, applying data augmentation, designing a CNN architecture, training and evaluating a model, and saving the trained model for inference.

Overall, the project helped strengthen foundational deep learning skills and provided hands-on understanding of an end-to-end image classification workflow suitable for internship-level AI tasks.


colab.research.google.com/drive/1IEfL8IKOVbweys-uu1n7dE_w_Y8r1zZT#scrollTo=tXdeSDna6q1N
Task2_DeepLearning_Image_Classification_CNN_CatsVsDogs
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Epoch 3/5
250/250 16s 64ms/step - accuracy: 0.6510 - loss: 0.6222 - val_accuracy: 0.6810 - val_loss: 0.5999
Epoch 4/5
250/250 21s 67ms/step - accuracy: 0.6702 - loss: 0.6078 - val_accuracy: 0.6755 - val_loss: 0.6029
Epoch 5/5
250/250 21s 70ms/step - accuracy: 0.6883 - loss: 0.5867 - val_accuracy: 0.6970 - val_loss: 0.5808
Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Training and Validation Loss
Training Loss
Validation Loss
63/63 1s 22ms/step - accuracy: 0.6925 - loss: 0.5754
Test Accuracy: 0.6925
Test Loss: 0.5754
Classification Report:
precision recall f1-score support
cat 0.73 0.62 0.67 1000
dog 0.67 0.77 0.71 1000
accuracy 0.69 2000
macro avg 0.70 0.69 0.69 2000
weighted avg 0.70 0.69 0.69 2000
Variables Terminal
02:39 Python 3
26°C Partly cloudy
Search
ENG IN
2:40 AM 6/28/2026


colab.research.google.com/drive/1IEfL8IKOVbweys-uu1n7dE_w_Y8r1zZT#scrollTo=tXdeSDna6q1N
Task2_DeepLearning_Image_Classification_CNN_CatsVsDogs
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Validation Accuracy
63/63 1s 22ms/step - accuracy: 0.6925 - loss: 0.5754
Test Accuracy: 0.6925
Test Loss: 0.5754
Classification Report:
precision recall f1-score support
cat 0.73 0.62 0.67 1000
dog 0.67 0.77 0.71 1000
accuracy 0.69 2000
macro avg 0.70 0.69 0.69 2000
weighted avg 0.70 0.69 0.69 2000
Confusion Matrix
619
381
234
766
Actual
Predicted
cat
dog
Model saved as /content/task2_simple_cnn_cats_dogs.keras
True: cat
Pred: cat
True: cat
Pred: cat
True: dog
Pred: dog
Variables Terminal
02:39 Python 3
26°C Partly cloudy
Search
ENG IN
2:41 AM 6/28/2026


colab.research.google.com/drive/1lEfL8IKOVbweys-uu1n7dE_w_Y8r1zZT#scrollTo=tXdeSDna6q1N
Task2_DeepLearning_Image_Classification_CNN_CatsVsDogs
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
RAM Disk
Predicted
Model saved as /content/task2_simple_cnn_cats_dogs.keras
True: cat
Pred: cat
True: cat
Pred: cat
True: dog
Pred: dog
True: dog
Pred: dog
True: dog
Pred: dog
True: dog
Pred: dog
Saved model loaded successfully.
Variables
Terminal
02:39
Python 3
26°C
Partly cloudy
Search
ENG
IN
2:41 AM
6/28/2026